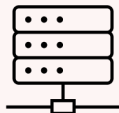


(a) Computing Bottleneck



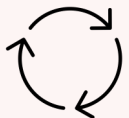
Extremely time-consuming

Replaying vast numbers of low-impact micro-events before reaching decisive states.



Requires massive resources

Typically needs supercomputers with thousands of cores and long wall-clock time.



Compute wasted on low-impact events

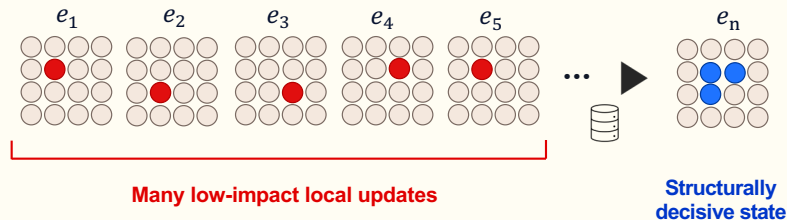
Simulation budget is consumed at micro-event resolution before reaching decisive state.

Compute is trapped in meaningless event replay instead of advancing along the true evolutionary backbone.

(b) Traditional Event-by-Event Simulation

Event-by-event (micro-resolution) advancement

The simulator advances through every micro-event sequentially.



○ Host atom

● Mobile atom

● Cluster



Supercomputers



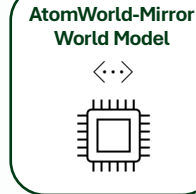
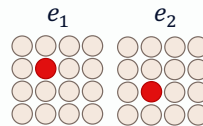
Weeks

(c) AtomWorld-Mirror: Macro-step World Modeling

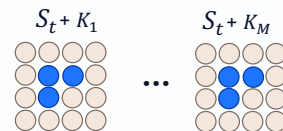
Learn from micro-evolution and predict Multi-K macro-steps

The world model predicts key events K steps ahead (multi-K macro-step)

Observe short micro-evolution



Predict Multi-K macro-steps



1*A100 GPU



Hours